ForeTac: Steering Robot Actions with Predicted Contact Consequences

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***Abstract*— Contact-rich manipulation requires decisions about physical interaction before the relevant tactile evidence is fully observed. A visually plausible motion can still lose contact during wiping, overload a fragile object, collide with a slot edge, or jam during insertion. This paper presents ForeTac, a framework for steering robot action chunks with predicted contact consequences. A base Diffusion Policy proposes an action chunk from visual, proprioceptive, and tactile observations. An action-conditioned tactile foresight model predicts the future GelSight marker-field latent sequence induced by the candidate chunk. A contact-quality energy model then scores the predicted tactile future, and its gradient is used for bounded trust-region refinement during late denoising. The result is a modular inference-time contact signal: the base policy remains the behavioral prior, while predicted tactile consequences provide a proactive correction before execution. We instantiate ForeTac on five real-robot contact tasks: board wiping, vase wiping, card swiping, chip grasping, and socket insertion. Current experiments validate the tactile latent representation, multi-horizon foresight accuracy, and bounded guidance gradients on blackboard wiping. We also report a 20-trial peg-in-hole insertion benchmark, where the foresight-guided policy reduces peak force and recovery time relative to raw DP and tactile concatenation.**

Index Terms–Contact-rich manipulation, tactile sensing, diffusion policy, tactile foresight, inference-time guidance.

I. INTRODUCTION

The central difficulty in contact-rich robot manipulation is that the outcome of an action is not fully determined by its geometric path. The same end-effector trajectory can be successful, ineffective, or unsafe depending on contact pressure, shear, impact, slip, and the temporal stability of those quantities. This distinction is easy to miss in tasks that look visually simple. In board wiping, a motion can cover the intended region while applying too little force to clean the surface or too much force for safe execution. In vase wiping, the same issue is compounded by a curved and fragile object. In card swiping, success requires sustained sliding contact through a narrow slot rather than merely reaching the slot entrance. In chip grasping, a small increase in pressure can break the object. In socket insertion, a small lateral contact before alignment can create bounce, jamming, or repeated retries. These examples share a common structure: the robot must choose an action based on the contact consequence it will create, not only on the current visual state.

Recent imitation-learning policies have substantially improved the action generation side of robotic manipulation. Action Chunking Transformers and Diffusion Policy learn expressive multi-step action distributions from demonstrations and can produce smooth behavior over long horizons [1], [2]. Their strength, however, also reveals a limitation. A policy trained to imitate successful trajectories may still make small local contact mistakes at deployment, especially when the object pose, surface normal, friction, or initial contact condition differs from the demonstrations. Adding tactile observations to the policy can improve the current state estimate [3]–[5], but standard tactile fusion remains mostly reactive: the robot observes contact after it happens and then changes future actions. For delicate contact, the critical decision often comes earlier. Before executing an action chunk, the robot should ask: if this candidate action is taken, what tactile state will it produce, and is that future contact desirable?

This question motivates a research gap between three lines of work that have largely remained separate. First, tactile representation learning provides compact encodings of high-dimensional touch signals [6], [7]. These representations improve perception, but by themselves do not decide how a candidate action should change. Second, visuo-tactile or visual world models predict future sensory states [8]–[11]. Prediction is useful, but many systems use it as an auxiliary training signal, a latent feature, or a planning model without an explicit differentiable link from predicted tactile quality back to the sampled action. Third, inference-time guidance and energy-based action selection can steer diffusion policies or rerank candidates [12]–[15]. Yet contact-rich manipulation needs a specific kind of guidance: the score should be grounded in the tactile future that the action itself is predicted to cause, and the update must stay close to the demonstrated behavior manifold so that the scorer cannot exploit out-of-distribution actions.

ForeTac is designed to close this gap. The framework treats future tactile consequences as an action-quality signal. At each control step, a base Diffusion Policy proposes an action chunk. A tactile foresight transformer then predicts a future sequence of GelSight marker-field latents conditioned on the candidate action. A contact-quality energy model scores the predicted future for task-appropriate contact. During selected late denoising steps, ForeTac backpropagates this score through the foresight model to the action sample and applies a small trust-region update. This converts tactile foresight from a passive prediction target into an active inference-time correction.

The design intentionally separates the roles of the modules. The diffusion policy is responsible for producing actions on the demonstration manifold. The foresight model is responsible for answering the counterfactual contact question: what tactile sequence is likely under this action? The energy scorer is responsible for distinguishing useful contact from task-specific failure modes such as contact loss, overload, pressure instability, slip, bounce, and jamming. The trust-region update is responsible for making only bounded local changes. This modularity is important academically and practically: it turns contact quality into an explicit testable object, allows independent evaluation of representation, prediction, scoring, and guidance, and avoids conflating better policy training with better inference-time contact reasoning.

This paper makes the following contributions.

- We formulate contact-rich action selection as prediction-conditioned guidance: actions are evaluated by the tactile consequences they are predicted to create, rather than only by their current observation likelihood.
- We introduce an action-conditioned multi-step tactile foresight module over compact TacVAE marker-field latents, preserving the temporal information needed to detect contact dropout, pressure spikes, and unstable force.
- We use a contact-quality energy model trained with semantic failure labels and deploy its margin score as a differentiable guidance signal for late-stage diffusion denoising.
- We define a five-task real-robot evaluation suite and report current evidence for tactile reconstruction, foresight prediction, bounded guidance diagnostics, and a completed peg-in-hole insertion benchmark, while explicitly separating these results from the remaining paired online rollout study.

II. RELATED WORK

A. Action-Chunk Policies for Robot Manipulation

Action-chunk policies reduce compounding errors by predicting a sequence of future controls rather than a single next action. ACT uses a transformer policy to predict smooth action chunks from demonstrations [1]. Diffusion Policy models the action chunk distribution with conditional denoising and has become a strong visuomotor imitation baseline [2]. Follow-up work improves 3D representations and sampling efficiency [16], [17]. Recent VLA policies such as $\pi_{0.5}$ further combine large-scale visual-language pretraining with robot action generation for open-world manipulation [18]. ForeTac is compatible with these policies because it does not replace the base action generator. It adds a local contact-quality objective during inference.

B. Tactile Representation and Prediction

Vision-based tactile sensors reveal local deformation, shear, and pressure that are difficult to infer from RGB images alone [3]–[5]. Learned tactile representations such as Sparsh and T3 provide transferable features for touch [6], [7]. Visuo-tactile world models and tactile prediction methods further learn how touch evolves under actions [9]–[11], [19]. Our focus is not only to predict touch, but to use predicted touch as an explicit differentiable action-quality signal. This distinction matters: recent world-model work treats future tactile prediction as a representation or planning aid, whereas ForeTac uses the predicted tactile future itself as the optimization target during inference.

C. Inference-Time Guidance and Candidate Verification

Energy-based imitation and candidate verification methods improve robot action selection by scoring or steering samples [12], [14]. DynaGuide uses dynamics-based guidance for diffusion policies [13], and touch-based guidance explores inference-time tactile steering [15]. ForeTac differs in the source of the score. TouchGuide is reactive to the current tactile state, while ForeTac scores the future tactile consequence that the current candidate action is predicted to create. The score is injected through a bounded update rather than selecting a full trajectory only after sampling.

III. PROBLEM FORMULATION

At time t, the robot observes

$$\mathbf{o}_t = \{I_t^g, I_t^w, q_t, \mathbf{m}_{t-W+1:t}\}, \tag{1}$$

where I_t^g and I_t^w are global and wrist images, $q_t \in \mathbb{R}^{d_q}$ is the robot state, and $\mathbf{m}_{t-W+1:t}$ is a recent tactile marker history. The base policy generates an action chunk

$$\mathbf{a}_{t:t+H-1} \in \mathbb{R}^{H \times d_a}. \tag{2}$$

The goal is to choose a chunk that both remains likely under the demonstration policy and produces high-quality future contact.

Let E_{tac} encode tactile marker histories into latent maps and let f_ψ be the action-conditioned tactile foresight model:

$$\hat{\mathbf{z}}_{t:t+H_f-1} = f_\psi(\mathbf{o}_t, \mathbf{a}_{t:t+H_f-1}). \tag{3}$$

A contact-quality scorer returns

$$s = S_\phi(\mathbf{a}_{t:t+H_f-1}, \hat{\mathbf{z}}_{t:t+H_f-1}, \tau), \tag{4}$$

where τ denotes the task or contact-quality semantics. ForeTac locally improves the diffusion action prior with the predicted-contact objective:

$$\log p_\theta(\mathbf{a}_{t:t+H-1} | \mathbf{o}_t) + \lambda S_\phi(\mathbf{a}, f_\psi(\mathbf{o}_t, \mathbf{a}), \tau). \tag{5}$$

The second term is not optimized freely. It is used as a small late-denoising correction so that the base policy remains the main behavioral prior.

IV. METHOD

A. System Overview

ForeTac has four stages. First, a TacVAE compresses tactile marker histories into structured latent maps. Second, a base Diffusion Policy learns the action prior from demonstrations. Third, a foresight transformer predicts future tactile latents from the current observation and candidate action chunk. Fourth, a contact-quality energy model scores the predicted tactile future and supplies an inference-time guidance gradient.

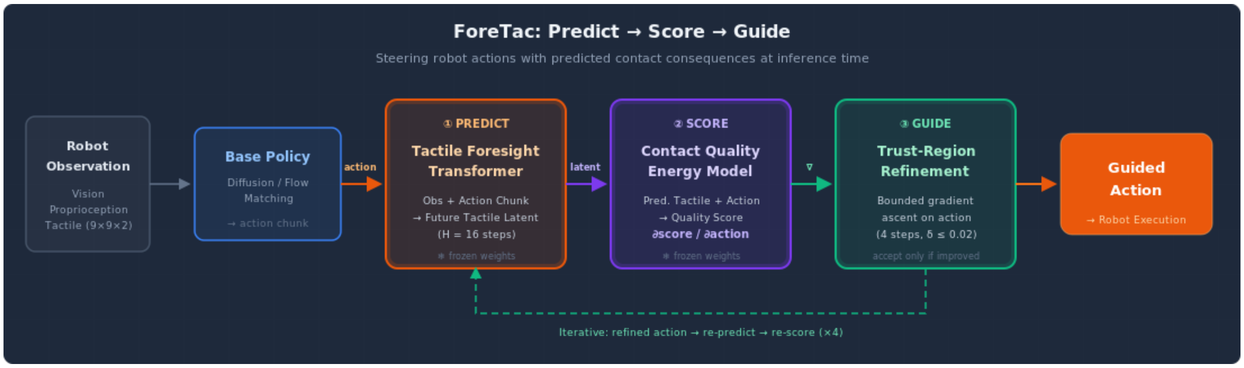

Fig. 1. **ForeTac overview.** The robot predicts future tactile consequences of candidate action chunks, scores the predicted contact quality, and applies a bounded correction before execution. The same predict-score-guide structure applies across wiping, swiping, grasping, and insertion tasks.

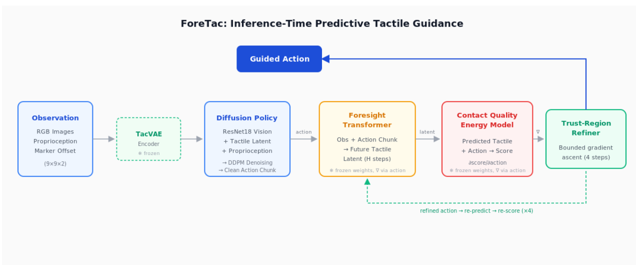

Fig. 2. **Architecture.** A base Diffusion Policy proposes an action chunk. Tactile foresight predicts the contact consequence of the chunk. A contact-quality energy model scores the prediction and sends a bounded gradient back to the action sample.

B. TacVAE Marker-Field Latents

The GelSight marker field is represented as a $9 \times 9 \times 2$ displacement grid. TacVAE encodes a recent marker history into a compact spatial latent

$$\mathbf{z}_t = E_{\text{tac}}(\mathbf{m}_{t-W+1:t}) \in \mathbb{R}^{16\times3\times3}. \tag{6}$$

The latent is flattened to 144 dimensions for foresight and scoring. The decoder maps latents back to marker displacement fields for reconstruction and prediction diagnostics. This representation preserves spatial contact structure while avoiding direct prediction in a noisy raw marker space.

C. Action-Conditioned Tactile Foresight

The foresight model predicts the tactile sequence induced by a candidate action chunk:

$$\hat{\mathbf{z}}_{t:t+H_f-1} = f_\psi(\mathbf{z}_t, q_t, \mathbf{a}_{t:t+H_f-1}). \tag{7}$$

The current implementation uses $H_f = 16$ for the main board setting. Predicting a sequence rather than a single endpoint is important because contact quality is temporal: a transient pressure spike, contact loss, or oscillation can make an otherwise plausible trajectory unsafe.

Training uses future TacVAE latents as targets. The loss combines latent prediction, decoded-marker reconstruction, and temporal-difference consistency:

$$\mathcal{L}_{\text{fs}} = \mathcal{L}_{\text{latent}} + \lambda_m \mathcal{L}_{\text{marker}} + \lambda_\Delta \mathcal{L}_\Delta. \tag{8}$$

For stride-aligned deployment, action and tactile target indices are sampled with the same temporal stride so that training and serving contracts match.

D. Contact-Quality Energy

The contact-quality scorer converts the predicted tactile future into a scalar score. Rather than hand-coding one force threshold, the scorer is trained with semantic contact labels. For board wiping, the labels are expert contact, pressure too small, pressure too large, and pressure unstable. For insertion, negative modes include pre-bounce, bounce, edge contact, and jamming. For vase wiping, card swiping, and chip grasping, the same structure is reused with task-specific labels such as contact dropout, slip, overload, and excessive object disturbance.

The deployed score is an unsaturated logit margin:

$$s_{\text{margin}} = \ell_{\text{good}} - \log \sum_{c\in\mathcal{C}_{\text{bad}}} \exp(\ell_c), \tag{9}$$

where ℓ_c is the class logit for contact class c. The margin is preferred over a saturated good-contact probability because it provides usable gradients even when the classifier is confident.

E. Trust-Region Denoising Guidance

Diffusion Policy denoises an action sample x_i at diffusion step i. From the predicted noise $\epsilon_\theta(x_i, i, \mathbf{o}_t)$, the clean-action estimate is

$$\hat{x}_0 = \frac{x_i - \sqrt{1-\bar{\alpha}_i}\,\epsilon_\theta(x_i, i, \mathbf{o}_t)}{\sqrt{\bar{\alpha}_i}}. \tag{10}$$

At selected late denoising steps, ForeTac denormalizes $\hat{x}_0$, predicts future tactile latents, computes the contact-quality score, and backpropagates:

$$g_i = \nabla_{x_i} S_\phi(\hat{x}_0, f_\psi(\mathbf{o}_t, \hat{x}_0), \tau). \tag{11}$$

The sample is updated by a bounded normalized step:

$$x_i \leftarrow x_i + \eta \frac{g_i}{\|g_i\|_2 + \epsilon}, \tag{12}$$

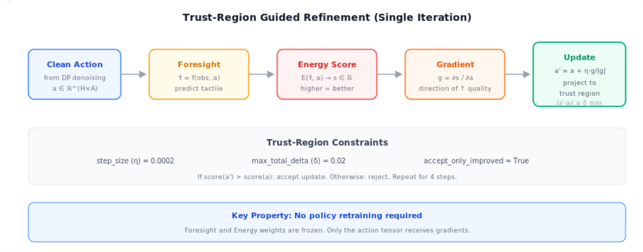

Fig. 3. **Guidance mechanism.** Guidance is applied only during late denoising, uses a normalized small step, and accepts bounded corrections that improve predicted contact quality.

followed by trust-region checks in raw action units. In the main latent-only guidance path, the action branch of the scorer is detached so that the cleanest gradient route is

$$\mathbf{a} \rightarrow \hat{\mathbf{z}}_{t:t+H_f-1} \rightarrow S_\phi. \tag{13}$$

This reduces the chance that the scorer exploits an action-only shortcut and keeps the guidance grounded in predicted tactile consequences.

V. EXPERIMENTAL SETUP

A. Tasks

We evaluate ForeTac on five contact-rich tasks. **Board wiping** requires stable moderate pressure while sweeping across a flat board. **Vase wiping** requires wiping a curved fragile surface without excessive normal force or object displacement. **Card swiping** requires continuous sliding contact and correct slot alignment. **Chip grasping** requires enough force to lift a fragile chip without crushing it. **Socket insertion** requires tight alignment while avoiding side contact, bounce, and jamming; we report a 20-trial peg-in-hole benchmark for this task below.

B. Baselines and Metrics

The full five-task paired rollout protocol compares ForeTac against a visual Diffusion Policy (DP), DP with tactile conditioning, a general VLA policy baseline, a $\pi_{0.5}$ VLA backbone, a reactive diffusion-transformer baseline, and a foresight autoregressive baseline. The primary metric is task success. The secondary metric is safe success, counted only when the task succeeds without unsafe contact, emergency stops, damaging interactions, or task-specific failure modes. Additional task metrics include force-band ratio, force variance, contact-dropout ratio, retry count, bounce count, and peak force. In this draft, the completed peg-in-hole insertion benchmark is reported below; the remaining task-level rollout matrix, including the $\pi_{0.5}$ evaluation, is still being collected under the same protocol.

C. Offline Evidence Required Before Rollouts

Before claiming online improvement, we require three checks. First, TacVAE must reconstruct marker fields accurately enough that downstream prediction preserves contact magnitude and direction. Second, tactile foresight must predict future latents and decoded marker fields across the horizons used by the controller. Third, the scorer must provide finite, bounded, and localized gradients when composed with the foresight model and the diffusion sampler. These checks do not replace real-robot success-rate comparisons, but they are necessary for safe and interpretable deployment.

TABLE I

TACVAE MARKER-FIELD RECONSTRUCTION QUALITY. CHIP EVALUATION IS PENDING.

Task	Norm. MAE ↓	Cosine ↑
Board wiping	0.039	0.998
Vase wiping	0.110	0.986
Card swiping	0.039	0.998
Chip grasping	TBD	TBD
Socket insertion	0.038	0.998

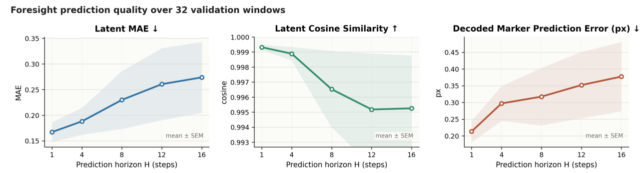

Fig. 4. **Foresight prediction quality.** Tactile foresight is evaluated by latent MAE, latent cosine similarity, and decoded marker-field error across horizons.

VI. CURRENT RESULTS

A. TacVAE Reconstruction

Table I reports the current TacVAE reconstruction evidence from the project page. Normalized MAE measures marker-field reconstruction error and cosine similarity measures directional agreement of the reconstructed tactile deformation.

B. Foresight Prediction Quality

The board foresight visualization evaluates horizons $H = \{1, 4, 8, 12, 16\}$ over 32 validation windows. Latent MAE increases from 0.167 at $H = 1$ to 0.274 at $H = 16$, while latent cosine remains high from 0.9993 to 0.9953. Decoded marker error increases from 0.213 px to 0.377 px across the same horizon range. These numbers indicate the expected horizon degradation while preserving the directional structure needed for contact-quality scoring.

C. Guidance Diagnostics

The current blackboard stride-3 diagnostic uses a temporal-stride-aligned foresight checkpoint and a latent energy scorer with expert-margin score. On the held-out validation split, the scorer separates expert, pressure-too-small, pressure-too-large, and pressure-unstable windows with accuracy 1.0, macro-F1 1.0, expert-vs-negative AUROC 1.0, and expert-margin AUROC 1.0 over 3144 windows. For 128 selected validation windows, all rows produce finite action gradients. The mean raw gradient norm is 2.4475 and the mean scaled norm with guidance scale 0.003 is 0.00734. Existing dense denoising logs show a mean guided-step score delta of 0.0198 with an applied update norm fixed near 0.003.

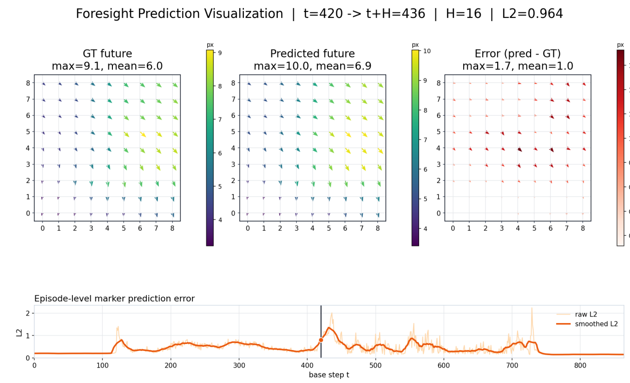

Fig. 5. **Predicted versus observed tactile consequence.** Example board-wiping visualization comparing predicted marker displacement fields with ground-truth future observations at the long horizon.

TABLE II

BLACKBOARD STRIDE-3 GUIDANCE DIAGNOSTIC. THIS IS AN OFFLINE GRADIENT/SAMPLER CHECK, NOT AN ONLINE SUCCESS-RATE RESULT.

Metric	Value
Validation windows	3144
Scorer accuracy / macro-F1	1.000 / 1.000
Expert-margin AUROC	1.000
Rows with finite gradients	128 / 128
Mean raw gradient norm	2.4475
Mean scaled gradient norm	0.00734
Mean guided-step score delta	0.0198
Mean applied update norm	0.0030

D. Peg-in-Hole Insertion Benchmark

Table III summarizes the completed 20-trial peg-in-hole study. Raw DP often overshoots contact, tactile concatenation improves success but still leaves high peak force, and the foresight-guided policies reduce both peak force and recovery time.

The average retry count measures bounce-lift-reinsert cycles. Recovery time is the elapsed time from peak force to force below 2 N. Raw DP never lifts after contact, so recovery time is undefined in those trials.

E. Generalist VLA Backbone Evaluation

Because ForeTac only assumes that the base policy proposes an action chunk, we also evaluate the same predicted-contact guidance on a $\pi_{0.5}$ VLA backbone. The local implementation appends tactile latent tokens to the $\pi_{0.5}$ prefix and keeps the tactile foresight branch as the contact consequence model. This experiment tests whether the contact-quality signal improves a generalist VLA backbone rather than only a task-local diffusion policy. Table IV is reserved for the completed rollout numbers.

VII. ABLATIONS AND FINAL EVALUATION PLAN

The final submission should include the following ablations. **No guidance versus ForeTac.** The same DP checkpoint should be run with guidance disabled and enabled so that the difference isolates inference-time contact guidance. **Reranking versus denoising guidance.** Best-of-K tactile

TABLE III

PEG-IN-HOLE INSERTION BENCHMARK WITH 20 TRIALS PER METHOD. DP+TAC DENOTES TACTILE CONCATENATION; DP+F DENOTES TACTILE FORESIGHT.

Method	Succ. ↑	Retry ↓	Max F (N) ↓	Peak F (N) ↓	Recover (s) ↓
Raw DP	35% (7/20)	0 (no lift)	28.1	24.3	–
DP+Tac	60% (12/20)	4.2	26.4	21.5	1.85
DP+F	85% (17/20)	1.1	15.2	12.8	0.42
RDT+F	90% (18/20)	0.9	14.1	11.5	0.35

TABLE IV

$\pi_{0.5}$ BACKBONE EVALUATION TEMPLATE. VALUES WILL BE FILLED AFTER THE CONTROLLED ROBOT TRIALS ARE COMPLETED.

Task	Metric	$\pi_{0.5}$	$\pi_{0.5}$+ForeTac	Δ
Board wiping	Safe succ. / force std.	TBD	TBD	TBD
Vase wiping	Safe succ. / peak force	TBD	TBD	TBD
Card swiping	Succ. / dropout ratio	TBD	TBD	TBD
Chip grasping	Safe succ. / peak force	TBD	TBD	TBD
Socket insertion	Succ. / recovery time	TBD	TBD	TBD

reranking tests whether the scorer can select better samples; denoising guidance tests whether the same score can locally improve the current sample. **Single-step versus multi-step foresight.** A single endpoint latent tests whether temporal contact failures can be captured without the full tactile sequence. **Score mode.** The expert margin in (9) should be compared with raw good-contact probability, clipped energy, and reason-aware energy. **Guidance path.** Action-only, tactile-only, and latent-only guidance separate scorer shortcuts from the intended action-to-foresight-to-score route. **Backbone transfer.** DP, RDT, and $\pi_{0.5}$ runs test whether the predicted-contact signal is tied to a particular action generator or transfers across chunk-based policy backbones.

VIII. DISCUSSION

ForeTac is motivated by a simple observation: current tactile feedback and future tactile consequence are different objects. Current tactile feedback is crucial after contact is observed, but many failures are already implied by the candidate action before the failure is visible. Predicting future touch creates a bridge from action choice to contact outcome.

The insertion benchmark matches this interpretation: raw DP and plain tactile concatenation still spend trials in bounce-lift-reinsert loops, while forethought-guided scoring lowers both peak force and recovery time.

The method also clarifies the role of inference-time optimization. Unbounded optimization of a learned contact score would be unsafe because the model could leave the demonstrated action distribution or exploit the scorer. ForeTac therefore uses the base policy as the dominant prior and restricts the contact score to small late-denoising corrections. The present diagnostics support that the gradients are finite and localized, but they do not by themselves prove task improvement. The decisive test remains paired real-robot evaluation using the same initial conditions and the same base policy weights.

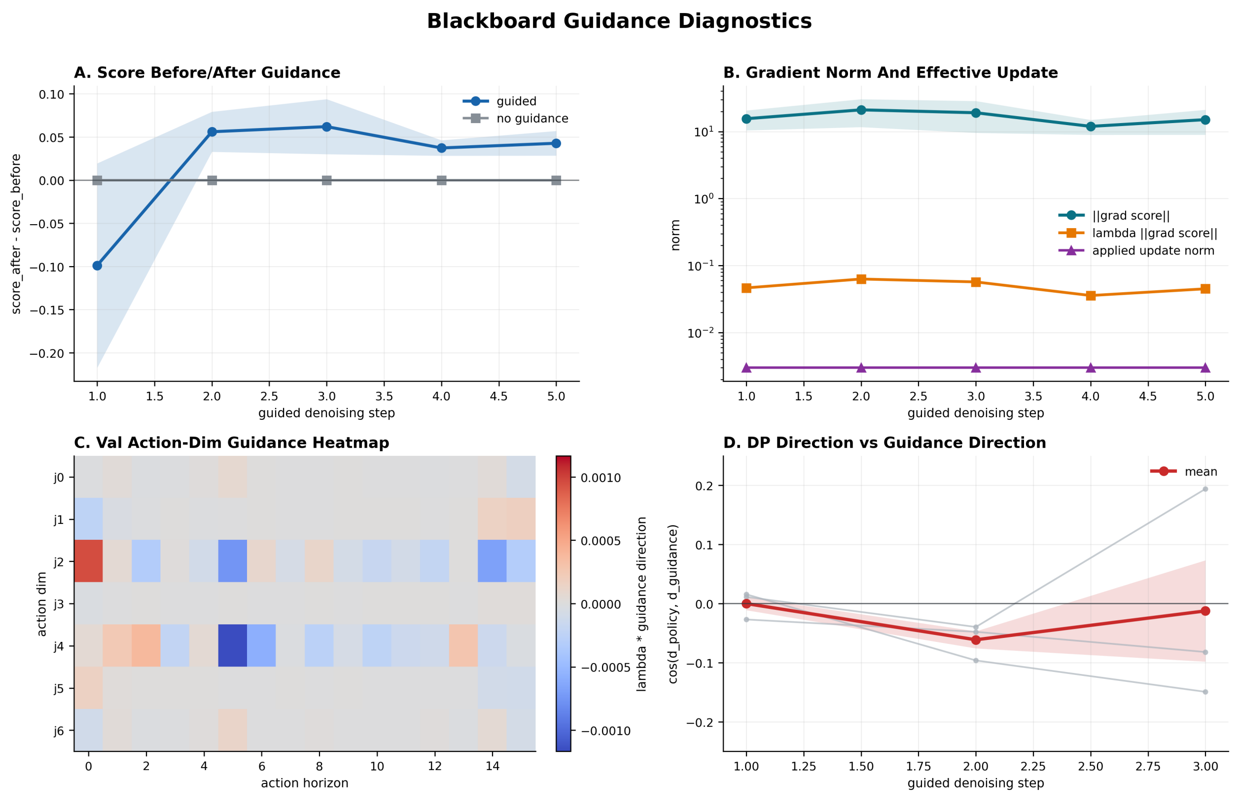

Fig. 6. **Guidance diagnostics for board wiping.** The diagnostic analyzes score changes, raw and scaled gradient norms, action-dimension localization, and policy-guidance alignment. The evidence supports that guidance acts as a small local correction rather than a global trajectory rewrite.

IX. Conclusion

We presented ForeTac, a framework for steering robot actions with predicted contact consequences. The method combines a base diffusion action prior, a TacVAE tactile latent representation, action-conditioned multi-step tactile foresight, and a contact-quality energy model whose gradient is applied through bounded late-denoising guidance. The paper reframes tactile prediction as an active action-quality signal: the robot predicts what contact an action will create and locally corrects the action before execution. Current results validate the representation, foresight, guidance pathway, and peg-in-hole insertion benchmark on the project tasks. Completing the remaining paired online success-rate matrix and ablation tables is the last requirement before final submission.

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